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The Directed Rural Postman Problem with Turn Penalties

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In this paper, we introduce a generalization of the directed rural postman problem including new features that can be encountered in practice when routes have to be operated on a street network: some turns are forbidden and other turns are allowed but with some penalties. This new problem is called the directed rural postman problem with turn penalties (DRPP-TP); we present some complexity results and three heuristics for the DRPP-TP: two of them are constructive, whereas the third one is an improvement heuristic. We also present a transformation of the DRPP-TP into an asymmetric traveling salesman problem (ATSP) that allows us to solve the problem exactly using an existing ATSP code. Computational results on a set of instances with up to 180 nodes and 666 arcs, are given.

Vehicle routing problems involve the design of efficient routes for a fleet of vehicles to satisfy the demands of customers located at different points. The practical importance of such problems is widely accepted, and many papers exist in the OR literature on this topic.

When routes have to be operated inside a city, they must observe traffic rules such as the prohibition of making certain turns, mostly left turns and U-turns. More generally, even if they are not forbidden, it is clear that certain turns may be dangerous for some trucks or more time consuming than others, so they should be penalized somehow. These issues have hardly been considered in the literature on vehicle routing despite their importance.

In this paper, we consider the directed rural postman problem with turn penalties (DRPP-TP), which consists of finding the minimum cost route that traverses all the arcs of a given set of required arcs in a directed graph. We consider two kinds of costs: the cost of traversing each arc and a penalty associated with each turn. Furthermore, certain turns are simply forbidden, so they cannot be used by any route.

If we do not take into account turn penalties and prohibitions in the DRPP-TP, we obtain the directed rural postman problem (DRPP), (CHRISTOFIDES et al., 1986; BALL and MAGAZINE, 1988; CAMPOS and SAVALL, 1995).

The shortest path problem with turn penalties has been studied, among others, by CALDWELL (1961), and KIRBY and POTTS (1969).

Problems similar to the DRPP-TP have been studied by BODIN and KURSH (1978, 1979) for routing street sweepers, BODIN et al. (1989) for vehicle routing in garbage collection, and ROY and ROUSSEAU (1989) in the design of postman routes. See also MCBRIDE (1982), which contains some ideas for decreasing the number of left and U-turns in a given route. Nevertheless, none of the algorithms developed in these works can guarantee to produce routes that avoid forbidden turns, they only consider turn penalties, and the algorithms try to avoid the forbidden turns by assigning them a high penalty. See also the recent surveys on arc routing problems by EISELT, GENDREAU, and LAPORTE (1995a,b) and by ASSAD and GOLDEN (1995). The latter reference stresses the importance of turn penalties and forbid-

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den turns in vehicle routing and contains an extensive discussion of the existing literature on this topic.

In this paper, we present some complexity results and three heuristics for the DRPP-TP: two of them are constructive, whereas the third is an improvement heuristic. We also present a transformation of the DRPP-TP into an asymmetric traveling salesman problem (ATSP), which allows the application of any existing exact or heuristic procedure for the ATSP to the DRPP-TP.

The paper is organized as follows. In Section 1, we formally define the DRPP-TP and study its complexity. Section 2 is devoted to the transformation of the DRPP-TP into an ATSP. We present, in Section 3, two constructive heuristics for the DRPP-TP, and, in Section 4, an improvement heuristic. Finally, Section 5 contains the computational results obtained with these heuristics and an exact procedure for the ATSP on two sets of instances: one containing randomly generated instances and the other containing real instances.

1. DEFINITIONS AND COMPLEXITY

THROUGHOUT THE PAPER, the word graph will refer to a directed graph. Otherwise, we will explicitly mention that the graph is non-directed. Given a graph $G = (V, A)$ and a node $v \in V$, a turn at v involves two arcs, say a and b, such that v is the head of arc a and the tail of arc b. The turn will be denoted $[a, b]$.

Consider a graph $G = (V, A)$ in which every arc $a \in A$ has associated a cost $c_a \geq 0$, a subset of turns is forbidden, and each allowed turn $[a, b]$ has associated a penalty $p_{ab} \geq 0$.

Given two arcs, a and b, we define a feasible chain from a to b as a sequence of arcs $C = \{a = a_1, a_2, \ldots, a_t = b\}$ such that, for $i = 1, \ldots, t - 1$, the head node of arc a_i is the tail node of arc a_{i+1} and the turn $[a_i, a_{i+1}]$ is allowed. The cost of the feasible chain C is defined as

$$c(C) = \sum_{i=1}^{t-1} (c_{a_i} + p_{a_i a_{i+1}}).$$

Note that the cost of the last arc is not considered in $c(C)$, but all the penalties of the turns that C makes are included in this cost.

We say that a feasible chain C is closed if $a = b$. Note that the penalty of the turn $[a_{t-1}, b]$ is included in $c(C)$ and the cost of arc a appears only once.

Let $G = (V, A)$ be a graph with:

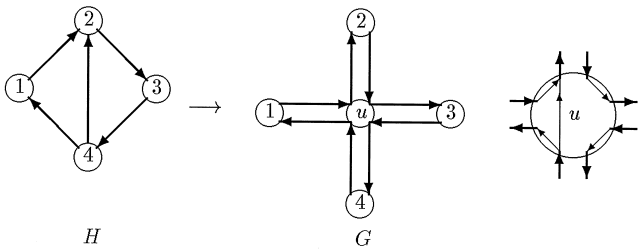


Fig. 1. Transforming the DHCP into a DRPP-TP. Drawing on the right shows the allowed turns at node u.

- a subset of forbidden turns, T_F,
- a nonempty subset of arcs, A_R, that will be called required arcs,
- nonnegative costs associated to its arcs, and
- nonnegative penalties associated to its allowed turns.

The DRPP-TP is defined as the problem of finding a minimum-cost feasible closed chain in G containing each required arc at least once. Any feasible closed chain that contains, at least once, every required arc will be called a feasible tour.

If all the turns are allowed and have zero penalties, the DRPP-TP reduces to the DRPP, which is an NP-hard problem in the strong sense (GAREY and JOHNSON, 1979). Therefore, the DRPP-TP is NP-hard in the strong sense. Nevertheless, the DRPP is solvable in polynomial time if all the arcs are required, although the following theorem shows that the DRPP-TP remains NP-hard in the strong sense even in this case.

THEOREM 1. *The DRPP-TP, defined on graph $G = (V, A)$ with a set of required A_R, is NP-hard in the strong sense even if $A_R = A$ and G is eulerian.*

Proof. We prove the theorem by polynomially transforming the directed Hamiltonian circuit problem (DHCP), a well known, strongly NP-complete problem, into the DRPP-TP. The DHCP is the following problem: given a directed graph $H = (N, E)$, decide whether there exists a circuit that traverses every node of H exactly once (Hamiltonian circuit). We transform the DHCP, defined on graph H, into a DRPP-TP defined on a graph $G = (V, A)$, where $V = N \cup \{u\}$, u being a new node, and $A = \{(v, u), (u, v): v \in V\}$ (see Fig. 1). All the arcs of A are required arcs and have the same cost, say b; a given turn $[(v, u), (u, w)]$ is a forbidden turn if $(v, w) \notin E$ or $v = w$ and all the allowed turns have zero penalties. Note that G is Eulerian and $A_R = A$.

Given a Hamiltonian circuit in H, we may associate it with a feasible DRPP-TP tour in G with cost $2|N|b$ as follows: for each arc (v, w) of the Hamilto-

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nian circuit, include a pair of consecutive arcs (v, u) and (u, w) in the DRPP-TP tour (note that the corresponding turn is allowed); given that the Hamiltonian circuit traverses each node exactly once, each required arc is traversed exactly once by the DRPP-TP tour, so the tour is feasible and its cost is $2|N|b$, the minimum possible cost. For instance, the Hamiltonian circuit (1, 2) (2, 3) (3, 4) (4, 1) of graph H, in Fig. 1, corresponds to the DRPP-TP feasible tour $(1, u)$ $(u, 2)$ $(2, u)$ $(u, 3)$ $(3, u)$ $(u, 4)$ $(4, u)$ $(u, 1)$ of G. The reverse transformation is obvious, given any feasible DRPP-TP tour with cost $2|N|b$, we may construct a Hamiltonian circuit of H by adding arc (v, w) to the circuit if, and only if, the pair of arcs (v, u) and (u, w) are consecutive, and appear in this order, in the DRPP-TP tour. Then, it follows that a Hamiltonian circuit exists in H if and only if the optimal solution of the DRPP-TP defined on G has cost $2|N|b$. □

Given that every section of a feasible tour is a feasible chain, the following proposition can be easily proved.

PROPOSITION 1. *The DRPP-TP defined on a graph $G = (V, A)$ has a feasible tour if, and only if, for every pair of required arcs a and b, there exist feasible chains in G from a to b and from b to a.*

A consequence of this theorem is that the feasibility of the DRPP-TP can be checked in time $O(p)$, where p is the number of allowed turns. The algorithm is a modification of the depth first search procedure (see, for instance, PAPADIMITRIOU and STEIGLITZ, 1982, chap. 9) that checks the connectivity of a directed graph. In our case, the algorithm is applied on a graph where the nodes represent the arcs of G and the arcs represent the allowed turns.

The algorithm of Dijkstra can be modified in a similar way to compute the shortest (minimum cost) feasible chain (s.f.c.) between two specified arcs. The complexity of the modified Dijkstra algorithm is $O(|A|^2)$. Note that the existence of forbidden and penalized turns implies that a shortest feasible chain may contain a node more than once. In fact, in a real route, it is quite usual that we are forced to make a loop to avoid a forbidden turn at a given intersection. However, a shortest feasible chain cannot contain any arc twice.

2. TRANSFORMING THE DRPP-TP INTO AN ATSP

THE FOLLOWING THEOREM shows that the DRPP-TP can be transformed into an ATSP.

THEOREM 2. *The DRPP-TP can be solved by transforming it into an ATSP defined on a graph with a number of nodes equal to the number of required arcs.*

Proof. Consider the DRPP-TP on a graph $G = (V, A)$ and let $A_R \neq \emptyset$ be the set of required arcs. For simplicity, we will suppose that the DRPP-TP is feasible in G. We construct a complete graph $G' = (V', A')$ where each node represents a required arc, and every arc $(u, v) \in A'$ has associated the cost of the s.f.c. from the arc corresponding to node u to the arc corresponding to node v.

Every Hamiltonian circuit H in G' can be transformed into a feasible tour T_H for the DRPP-TP by substituting each arc of H by the corresponding s.f.c. in G excluding its last arc. The cost of T_H $(c(T_H))$ is then the same as the cost of H in G', say $c'(H)$.

In contrast, if T is a feasible DRPP-TP tour in G, we can construct a Hamiltonian circuit in G' as follows: let $a_1, a_2, \ldots, a_m$ be the order in which the required arcs $a_i \in A_R$ $i = 1, \ldots, m$ are traversed for the first time in T (starting at any arbitrary node), then, visiting the corresponding nodes of G' in this order gives us a feasible solution, say H_T, for the ATSP. Note that the cost of the section of T from a_i to a_{i+1} including c_{a_i} but not $c_{a_{i+1}}$ is at least equal to the cost of the s.f.c. from a_i to a_{i+1}. Then, it holds that $c(T) \geq c'(H_T)$.

Let H^* be the optimal solution of the ATSP in G'. For every feasible tour T for the DRPP-TP,

$$c(T) \geq c'(H_T) \geq c'(H^*) = c(T_{H*}),$$

where T_{H*} is the DRPP-TP feasible tour obtained from H^*. Therefore, T_{H*} is an optimal solution for the DRPP-TP. □

There exists in the literature several methods to solve the ATSP, both exactly and heuristically (see, for instance, the survey of LAPORTE, 1992). Therefore, this transformation can be used as an easy way to solve the DRPP-TP. Nevertheless, the transformation has some drawbacks: the transformed graph is considerably larger than the original one and, furthermore, if we transform the problem into an ATSP, we cannot make use of the specific characteristics of the DRPP-TP.

For this reason, it is important to develop fast heuristics for the DRPP-TP that could be applied to large instances as they appear in practical situations.

The transformation into an ATSP also provides an easy way to compute a lower bound for the DRPP-TP by solving an assignment problem (AP), where the cost of assigning the required arc a to the required arc b is equal to the cost of the s.f.c. from a to b, except when $a = b$, in which case, the cost is con-

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sidered to be infinity. It is obvious that the optimal cost to this AP is a lower bound for the DRPP-TP.

The DRPP-TP can also be transformed into a DRPP (SOLER, 1995) but this transformation is only useful in the case where the number of allowed turns with positive penalties is small. Furthermore, the DRPP has not received nearly as much attention as the ATSP so we do not present this transformation here.

3. TWO CONSTRUCTIVE HEURISTICS FOR THE DRPP-TP

BODIN AND KURSH (1978, 1979), studied a problem similar to the DRPP-TP for which they proposed a heuristic algorithm. Later, Bodin et al. (1989), applied this heuristic, with some modifications, to solve a real problem in the town of Oyster Bay, New York. The main difference with the problem we address is that they did not consider the existence of forbidden turns. Furthermore, the nodes are supposed to be points in the plane with known coordinates. The Bodin and Kursh heuristic (abbreviated BK heuristic) uses a simple operation, called turn crossing, to join two circuits containing the same node, into a single circuit. Let $[a, b]$ be a turn made by circuit C_1 at node v and let $[c, d]$ be a turn made by circuit C_2 at the same node v. Crossing these two turns means substituting them by the turns $[a, d]$ and $[c, b]$, in such a way that circuits C_1 and C_2 become joined.

We describe below, with some detail, the BK heuristic because the first heuristic we propose for the DRPP-TP can be considered as an improvement of it and some steps are similar.

In what follows, we will assume that a very high penalty, M, is assigned to every forbidden turn.

BODIN AND KURSH HEURISTIC

Input: A DRPP-TP instance defined on a graph $G = (V, A)$

Step 1. Solve a minimum cost flow problem on graph G where the turn penalties are not taken into account and the supply/demand of every node v is s_v = (number of required arcs entering v) − (number of required arcs leaving v). Let A_F be a set of arcs containing as many copies of every arc of G as the flow units traversing it in the optimal solution of the minimum cost flow problem. Let $G' = (V, A_R \cup A_F)$ and denote by G'_F the graph induced in G' by the set of arcs $A_R \cup A_F$. Note that G'_F is symmetric but it may be nonconnected.

Step 2. For every node v of graph G'_F, solve an AP to assign, at minimum cost, every required arc entering v to a required arc leaving v, where the assignment cost is the corresponding turn penalty. Consider the arc assigned to arc a as the successor of a. Then, by recursively following each arc by its successor, we obtain a set of circuits that cover all arcs of G'_F. When translated to graph G, these circuits cover all the required arcs of G at least once. Then, if only one circuit exists, STOP, it is a solution for the DRPP-TP except for the fact that it may violate some forbidden turns.

Step 3. While there exist pairs of circuits having a node in common, join them by crossing the appropriate turns, until only one circuit exists in every connected component of G'_F. If only one component exists, STOP.

Step 4. Let $N_1, \ldots, N_t$ be the connected components of G'_F. Consider a complete nondirected graph $H = (V_H, E_H)$ where node $i \in V_H$ represents component N_i and the cost of edge $(i, j) \in E_H$, c_{ij} is defined as the minimum euclidean distance between any node in N_i and any node in N_j.

Step 5. Solve a shortest spanning tree problem (SST) on graph H. For each edge (i, j) of the SST, do:

—identify the nodes, say u and v, in components N_i and N_j, respectively, for which the minimum was attained when computing c_{ij},
—compute the shortest path in G (without taking into account turn penalties) from u to v and from v to u, and
—add the circuit formed by these two paths to G'_F and join it to the existing circuits in the connected components N_i and N_j by crossing the appropriate turns.

When all edges of the SST have been considered, STOP. A unique circuit has been obtained, which is a (possibly nonfeasible) solution to the DRPP-TP.

As has been already remarked, the solution given by the BK heuristic may violate some forbidden turns. We present here a method that guarantees the elimination of any forbidden turn made by the solution, provided that the DRPP-TP is feasible.

Let T be a tour traversing every required arc of G at least once. Let $a_1, a_2, \ldots, a_s$ be a section of T; if we delete from T the section $a_2, \ldots, a_{s-1}$ and the resulting chain still traverses every required arc at least once, we say that $a_1, a_2, \ldots, a_s$ is a nonrequired section of T. Note that, if $s = 2$, a_1, a_2 is always a nonrequired section of T. A nonrequired section of T is said to be maximal if it is not contained in a longer nonrequired section of T.

If $a_1, a_2, \ldots, a_s$ is a maximal nonrequired section of T then, necessarily, a_1 and a_s are required arcs.

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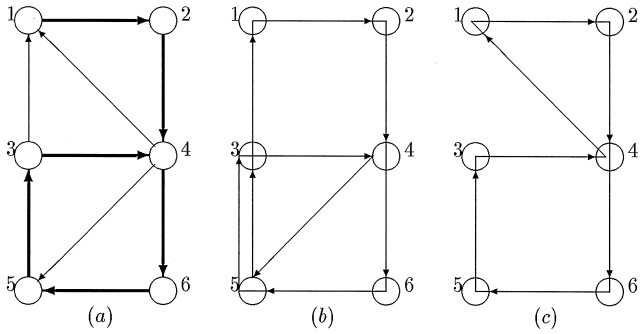


Fig. 2. (*a*) Graph G. Required edges: bold lines. (*b*) Tour violating forbidden turn [(4, 5), (5, 3)]. (*c*) Feasible tour. The maximal nonrequired section (3, 4), (4, 5), (5, 3), (3, 1), (1, 2) has been substituted by the shortest feasible chain between arcs (3, 4) and (1, 2): (3, 4), (4, 1), (1, 2).

If, for instance, a_s were not a required arc, let a_{s+1} be the arc following a_s in tour T, then $a_1, a_2, \ldots, a_s, a_{s+1}$ would also be a nonrequired section of T, so $a_1, a_2, \ldots, a_s$ would not be maximal.

If T violates a forbidden turn, say $[a, b]$, we can easily find a maximal nonrequired section of T containing the section a, b, say $a_1, \ldots, a_s$. By replacing this section by the shortest feasible chain between a_1 and a_s, we avoid the forbidden turn $[a, b]$. This shortest feasible chain between a_1 and a_s always exists, assuming that the DRPP-TP is feasible, because a_1 and a_s are required arcs (see Proposition 1).

As an example, consider the DRPP-TP defined on the graph depicted in Figure 2*a*, where required arcs are those drawn with bold lines and nonrequired arcs are those drawn with thin lines; the cost of every arc is 1, all the turn penalties are 0 and the only forbidden turn is [(4, 5), (5, 3)]. The tour T represented in Figure 2*b* covers all the required arcs but violates the forbidden turn. The procedure to eliminate this forbidden turn is: starting with the tour section (4, 5), (5, 3), we enlarge it by adding arcs to the first or last arc in the section while the tour section satisfies the conditions given in the definition of a nonrequired section. Thus, we obtain the maximal nonrequired section (3, 4), (4, 5), (5, 3), (3, 1), (1, 2) (note that the removal of section (4, 5), (5, 3), (3, 1) from tour T results in a chain that still contains each required arc at least once). Then we may substitute it by the shortest feasible chain between arcs (3, 4) and (1, 2), i.e. (3, 4), (4, 1), (1, 2), thus obtaining a new tour, depicted in Figure 2*c*, that does not violate the forbidden turn.

This procedure is used by our heuristic, CONS1, to avoid those forbidden turns that may appear while constructing a tour for the DRPP-TP and also by the improvement heuristic presented in Section 4.

Another step of the BK heuristic that can be improved is step 3, where circuits are joined by crossing turns made at the same node. These crossings introduce an extra cost into the solution that we have tried to minimize. McBride (1982), proposed the following strategies:

- First, join circuits with an arc in common. In this case, the extra cost is zero.
- Join, if possible, three circuits containing the same node. In this case, there are two possible pairs of crossings to join the three circuits into a single one and we can select the one producing the minimum extra cost.

We propose the following procedure, called CROSSING, to improve this step by considering the joining process as a whole. The procedure CROSSING is presented here in a more general setting than that of step 3 of BK algorithm because it will also be used in other heuristics for the DRPP-TP.

PROCEDURE CROSSING

Input: A symmetric and connected graph H (not necessarily simple), a set of circuits $C_1, \ldots, C_l$ that cover all the arcs of H exactly once and a thresholding value Q. Let $s(a)$ denote the successor of arc a in the circuit to which it belongs. For simplicity C_i will also denote the set of arcs in the circuit.

Step 1. Join any pair of circuits containing a copy of the same original arc, making a crossing that involves these copies, until no such pair of circuits exist. If only one circuit remains, STOP.

Step 2. Let $C_1, \ldots, C_t$ be the present set of circuits covering H. Build a nondirected and connected graph $H_C = (V_C, E_C)$ where node $i \in V_C$ represents circuit C_i, $i = 1, \ldots, t$ and $(i, j) \in E_C$ iff circuits C_i and C_j have at least one node in common. The cost associated with edge (i, j) is:

$$c_{ij} = \min\{p_{as(b)} + p_{bs(a)} - p_{as(a)} - p_{bs(b)}:$$

$$a \in C_i,\ b \in C_j,$$

and a and b are arcs entering in the same node}

Step 3. Compute the SST in H_C. If all the edges in the SST have a cost greater than Q, STOP.

Step 4. For every edge (i, j) of the SST, with cost $c_{ij} < Q$, do:

Let $a \in C_i$ and $b \in C_j$ the arcs for which the minimum is obtained when computing c_{ij}. If $s(a)$ and $s(b)$ have not changed as a result of preceding crossings made using the present SST, cross the turns $[a, s(a)]$ and $[b, s(b)]$, so

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the circuits C_i and C_j become joined, and update $s(a)$ and $s(b)$.

If more than one circuit remains, go to step 2. Otherwise, STOP.

Note that each turn crossing made at step 4 modifies the successors of two arcs, so subsequent crossings involving the same arc have an extra cost which does not correspond to the one computed in step 2. Then, it makes no sense to use any of them in another turn crossing, before the corresponding extra cost has been recomputed by executing steps 2 and 3 again.

We now present a constructive heuristic for the DRPP-TP that implements all these ideas. It uses the heuristic proposed by Ball and Magazine (1988), for the DRPP, as a first step to obtain a symmetric and connected graph without taking into account turn penalties. The Ball and Magazine heuristic consists basically of two steps.

1. Construct a symmetric graph containing at least one copy of every required arc of G, using the method explained in step 1 of the BK heuristic. Let G'_F denote this graph.
2. Connect the components of G'_F in a similar way to that explained in steps 4 and 5 of the BK heuristic, except for the way in which costs for the SST are computed. That is, for every pair of connected components, say A and B, of G'_F, compute the cost c_{AB} = min{sum of the costs of the shortest paths from u to v and from v to u: $u \in A$, $v \in B$}, then compute the SST using these costs and add to G'_F the pairs of paths corresponding to the edges of the SST. The result is a symmetric and connected graph.

HEURISTIC CONS1

Input: A DRPP-TP instance defined on graph $G = (V, A)$.

Step 1. Check the feasibility of the DRPP-TP instance (see Proposition 1). If the instance is not feasible, STOP.

Step 2. Apply the DRPP heuristic of Ball and Magazine to the DRPP-TP instance without taking into account turn penalties. Let G'_F be the resulting symmetric and connected graph, which contains at least one copy of each required arc of G.

Step 3. Partition G'_F into a set of circuits by solving an AP at each node as in step 2 of the BK heuristic. Let $C_1, \ldots, C_k$ be the circuits that cover G'_F. If $k = 1$, go to step 5.

Step 4. Apply procedure CROSSING to G'_F, circuits $C_1, \ldots, C_k$ and threshold value $Q = 3M$ (which means that a crossing may introduce a forbidden turn). The result is a unique circuit, say T, that may violate some forbidden turns.

Step 5. While T violates a forbidden turn, do:

Let $[a, b]$ be a forbidden turn made by T, find a maximal nonrequired section of T containing $[a, b]$, say $a_1, \ldots, a_s$, and replace this section by the shortest feasible chain from a_1 to a_s.

The following heuristic constructs a DRPP-TP solution by first solving an AP that involves all the required arcs at the same time instead of solving several APs, one at each node, as do the preceding heuristics. This AP is, in fact, the same that provides a lower bound for the DRPP-TP mentioned in Section 2. Using the AP solution, we construct a set of circuits covering all the required arcs at least once. Procedure CROSSING is then used with threshold value M/2 to join the largest possible number of circuits without introducing any violated forbidden turn. Finally, a modification of the CROSSING procedure (called CROSSING2 for short) is applied to ensure that all the remaining circuits are joined. CROSSING2 is like CROSSING except for the following features:

- The input graph may be nonconnected.
- In step 2, the cost associated with edge (i, j), c_{ij}, is computed as follows:

$$c_{ij} = \min\{g_{a,s(b)} - c_a + g_{b,s(a)} - c_b - p_{a,s(a)} - p_{b,s(b)}: a \in C_i, \quad b \in C_j\}$$

where $g_{x,y}$ denotes the cost of the shortest feasible chain in G from arc x to arc y.

- The joint of circuits C_i and C_j in step 4 is made by substituting the turns $[a, s(a)]$ and $[b, s(b)]$ by the shortest feasible chains from a to $s(b)$ and from b to $s(a)$ respectively, excluding their initial and final arcs.

Note that these modifications allow any two circuits to be joined with a cost less than M.

HEURISTIC CONS2

Input: A DRPP-TP instance defined on graph $G = (V, A)$.

Step 1. For every ordered pair (a, b), $a \neq b$, of required arcs of G, compute g_{ab} = cost of the shortest feasible chain from a to b. If $g_{ab} > M$ for any pair of required arcs a and b, STOP, the instance is not feasible. Otherwise, solve the AP defined by the matrix of assignment costs $[g_{ab}]_{a,b \in A_R}$ (where $g_{aa} = M$). For every $a \in A_R$, let $q(a)$ be the arc to which it has been assigned in the optimal AP solution.

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Step 2. Initialize graph $G' = (V, \varnothing)$. Until all the required arcs are covered, do:
  Select a required arc not yet covered, say $a = a_0$, do $i := 0$, and do:
    Add to G' the shortest feasible chain from a_i to $q(a_i)$ excluding arc $q(a_i)$. Mark a_i as covered and do $a_{i+1} := q(a_i)$, $i := i + 1$.
  until $a_i = a_0$ (a circuit has been formed).

If G' is connected and only one circuit has been generated, STOP, it is a feasible tour for the DRPP-TP.

Step 3. Apply procedure CROSSING to every connected component of G' and its corresponding set of circuits, with threshold value $Q = M/2$ (that is, only crossings that do not introduce forbidden turns are allowed). If only one circuit remains, STOP, it is a DRPP-TP feasible tour.

Step 4. Let $C_1, \ldots, C_r$ be the present set of circuits. Apply procedure CROSSING2 to this set of circuits, with threshold value $Q = M$. The result is a single circuit that does not violate any forbidden turn, so it is a feasible tour for the DRPP-TP.

Note that the optimal cost of the AP solved in Step 1 is a lower bound for the DRPP-TP, then, if it is equal to the cost of the heuristic solution CONS2 produces, this will be optimal.

4. AN IMPROVEMENT HEURISTIC

WE PRESENT in this section an improvement heuristic based on the use of two operations that locally modify a given tour solution of the DRPP-TP. One operation consists of the substitution of a maximal nonrequired section of the tour by the appropriate shortest feasible chain. The other operation, called 3-interchange, consists of crossing three appropriate turns made by the tour at the same node to obtain a new tour that differs from the original only in the turns made at this node.

Let C be a circuit passing at least three times through a given node v and let $s(a)$ denote the successor of arc a in C. Let a_1, a_2, and a_3 be three arcs entering node v that are traversed by circuit C in that order, that is, circuit C can be written $a_1 s(a_1) S_1 a_2 s(a_2) S_2 a_3 s(a_3) S_3$, where S_1, S_2, and S_3 are sections of C. A 3-interchange associated to arcs a_1, a_2, and a_3 means substituting the three turns made by C at v, $[a_1, s(a_1)]$, $[a_2, s(a_2)]$, and $[a_3, s(a_3)]$, by turns $[a_1, s(a_2)]$, $[a_2, s(a_3)]$, and $[a_3, s(a_1)]$, which we assume to be not forbidden. The result is a new circuit that traverses the same set of arcs in a different order $a_1 s(a_2) S_2 a_3 s(a_1) S_1 a_2 s(a_3) S_3$ (see Fig. 3). The cost of the new circuit changes by the quantity $p_{a_1 s(a_2)} + p_{a_2 s(a_3)} + p_{a_3 s(a_1)} - p_{a_1 s(a_1)} - p_{a_2 s(a_2)} - p_{a_3 s(a_3)}$, that will be called cost increment of the 3-interchange.

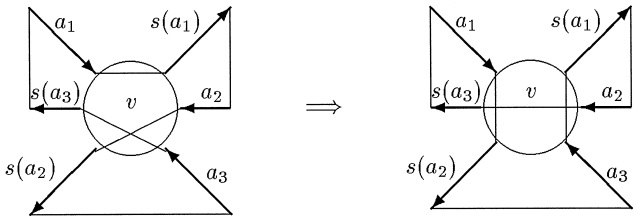


Fig. 3. Example of 3-interchange in a node v.

Note that it is necessary to know the order in which the entering arcs a_1, a_2, and a_3 are traversed by C because, otherwise, the circuit could be split into three circuits by the 3-interchange.

PROCEDURE IMPROVE

Input: A solution tour T of the DRPP-TP, R = maximum number of iterations without improvement in Step 2, and m = cardinality of the groups of nodes used in step 4.

Step 0. $H := T$ (H represents the current best solution found), $k := 0$.

Step 1. Label each arc contained in T.

Step 2. While a labeled arc a exists, a being a nonrequired arc or a required arc such that at least another copy of it is also labeled, do:

—Search for a maximal nonrequired section of T containing arc a and replace it in T by the shortest feasible chain between its end arcs.
—Remove the labels of the arcs corresponding to the maximal nonrequired section that has been replaced.
—If $c(T) < c(H)$, do $H := T$ and $k := 0$.

Step 3. (The 3-interchanges are made on tour T in this step.) If $k = R$ go to step 4. Otherwise do $k := k + 1$ and for every node v visited by tour T do:
  For every arc of T entering v, search the 3-interchanges containing it. If one 3-interchange is found with a negative cost increment, do it and update H. Otherwise do any of them (without taking into account its cost increment), if any exists.
If no 3-interchange has been made in any node, STOP. Otherwise go to step 1.

Step 4. (The 3-interchanges are made on tour H in this step.) Group the nodes visited by tour H into subsets of m elements: $L_1, \ldots, L_q$ (obviously L_q could contain less than m nodes).
Repeat:

  For every group L_i, $i = 1, \ldots, q$, do:
  Explore all the 3-interchanges that can be made

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in any node of set L_i, and select the one with minimum cost increment. If this cost increment is negative, make the 3-interchange in H.

Until no 3-interchange has been made, in which case, STOP.

Note that the 3-interchanges are made in two different steps of the algorithm and with different purposes. Usually, the largest decreases in cost are achieved in step 2 by substituting a maximal nonrequired section by the corresponding shortest feasible chain. Then, the objective of the 3-interchanges made in step 3 is to modify the solution tour, to give new opportunities in the search for maximal nonrequired sections, because changing the order in which the arcs of the tour are traversed may result in different maximal nonrequired sections.

The objective of step 4 is to improve the solution as much as possible using only 3-interchanges. Accordingly, only 3-interchanges with negative cost increment are made on tour H. To efficiently explore possible 3-interchanges, we compute, for each arc a, the label $n(a)$, which is the next arc entering the same node that a enters, when the tour is traversed starting at a. Using these labels, it is easy to explore all the 3-interchanges at a given node. Nevertheless, the labels have to be updated every time a 3-interchange is made. For this reason, we try to limit the number of 3-interchanges made in step 4 by performing the best one among those possible 3-interchanges at a subset of m nodes. We have tried with different values of m and the experience showed that small values of m imply more time taken to update labels $n(a)$, whereas a large value of m means taking up a lot of time just exploring 3-interchanges. A value that balances these effects is $m = 5$, which has been adopted in the computational experience we present in the next section. The maximum number of iterations without improvement in Step 2, R, has been set equal to 3, because larger values produced almost the same results but with larger computational times.

In future research, it could be possible to incorporate within the procedure IMPROVE the ideas of modern metaheuristics: tabu search, simulated annealing, genetic algorithms, etc. These techniques usually produce near optimal solutions, but with large CPU times. As we will show in the next section, our heuristics usually produce solutions with cost near to the lower bound, with small CPU times, so it seems that, at least for the set of instances we have tried, it is not worthwhile to apply more time-consuming techniques.

5. COMPUTATIONAL RESULTS

WE HAVE CONSTRUCTED two sets of instances to test the heuristic algorithms described above. The first set contains 24 randomly generated instances, with up to 180 nodes and 666 arcs, that are intended to resemble the street network of a city with mostly two-way streets.

These instances are defined on grid graphs that have been generated as follows. Let n and m be, respectively, the numbers of rows and columns of the grid. We randomly select n points in the plane among those in the set $\{(0, 1), (0, 2), \ldots, (0, 1000)\}$ and m points in the set $\{(1, 0), (2, 0), \ldots, (1000, 0)\}$. Let $(x_1, 0), \ldots, (x_m, 0)$ and $(0, y_1), \ldots, (0, y_n)$ be the selected points. Then, consider the directed grid graph whose node set is $V = \{p_{ij} = (x_i, y_j)$: $i = 1, \ldots, m, j = 1, \ldots, n\}$ and whose arc set is $A = \{(p_{ij}, p_{kl})$: $i = k$ and $|l - j| = 1$, or, $j = l$ and $|i - k| = 1\}$. The cost of each arc is equal to its length (which will be an integer). Each arc is declared required with a given probability; similarly, a given percentage of left turns and a given percentage of U-turns are declared forbidden. Finally, each allowed turn has a penalty assigned according to the type of turn: left, right, U-turn, and go straight. The penalties assigned to each kind of turn are shown in Table I for each instance. If one generated instance was infeasible, we simply rejected it.

Some cities have areas where most of the streets are one-way streets. It is more difficult to generate random graphs resembling this kind of street network, so we have decided to use the real street networks of some Spanish cities. Thus, instances T1 to T5 represent 5 different zones of Torrente, and V represents a zone of Valencia. The forbidden turns are the real ones and the turn penalty assigned to each allowed turn is shown in Table II. Nevertheless, if a real "give way" or "stop" signal existed at a given turn, its penalty has been increased in one or two units, respectively.

All the codes were written in FORTRAN and run on a Sun Sparcstation 20.

Table III presents the computational results obtained on both sets of instances by the heuristic algorithms: the BK heuristic, our two constructive heuristics (CONS1) and (CONS2), and the improvement heuristic (IMP). It also includes the results obtained with an exact procedure that consists of transforming each instance into an ATSP instance, and running the branch-and-bound code of FISCHETTI and TOTH (1992), which solves the ATSP exactly. This code was run with a CPU time limit of 3 hours. Table III shows the optimal cost of those

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TABLE I

Characteristics of the Random Instances

				% Forbidden Turns		Turn Penalty			
	$n \times m$	$\|A\|$	$\|A_R\|$	Left	U	Right	Left	U	Straight
P1	5×7	116	116	15	30	2	4	8	1
P2	6×6	124	79	20	50	3	5	11	1
P3	4×10	132	98	10	40	3	5	11	1
P4	7×7	168	148	20	50	1	3	7	1
P5	7×7	168	128	20	60	2	3	7	1
P6	6×9	186	166	15	30	2	4	8	1
P7	6×10	208	131	15	30	2	4	8	2
P8	12×8	344	212	15	60	2	4	8	0
P9	12×8	344	344	25	50	2	4	8	0
P10	9×11	356	176	10	100	2	3	–	1
P11	10×10	360	360	20	75	3	5	11	1
P12	7×15	376	376	20	40	1	4	8	2
P13	7×15	376	177	20	80	1	4	8	1
P14	9×14	458	354	15	40	3	5	11	2
P15	9×14	458	458	10	90	3	5	11	2
P16	13×11	524	358	10	90	3	5	11	2
P17	13×11	524	524	10	90	3	5	11	2
P18	12×12	528	377	5	75	1	4	8	2
P19	12×12	528	427	10	50	1	4	8	2
P20	10×15	550	425	20	50	3	5	11	1
P21	10×15	550	550	20	50	3	5	11	1
P22	13×13	624	556	25	80	1	3	8	1
P23	13×13	624	451	15	80	1	3	8	1
P24	15×12	666	540	15	100	2	5	–	1

instances that could be optimally solved within this time limit, and a lower bound for the remaining instances.

Fischetti and Toth (1992), claim that their ATSP code performs very well on instances with very asymmetric costs (i.e., cost c_{ij} is very different from c_{ji}), while it is not so effective on instances with almost symmetric costs. Our experience confirms this; the randomly generated instances involve almost symmetric costs when transformed into ATSP instances, and 8 out of 24 could not be optimally solved, while the city instances have highly asymmetric costs and all of them were optimally solved.

Our implementation of the BK heuristic slightly differs from the heuristic described by Bodin and Kursch (1978, 1979): costs c_{ij} in Step 4 are the costs of the corresponding shortest paths in graph G instead of the Euclidean distances, and the circuit joining of Step 3 and 5 is made trying to avoid forbidden turns except when no other possibility existed. Despite this fact, the solutions obtained by the BK heuristic were not feasible in 18 out of the 30 instances tried. The costs of these solutions, as shown in Table III, include the high penalty M assigned to each violated forbidden turn. Thus, a cost of $22011 + 4M$ indicates that the solution is infeasible because it violates 4 forbidden turns. Note that BK produced the best solutions in two instances: P11 and P15.

Tables IV and V summarize the results obtained by the different heuristics. The deviation is computed as $100 * (\text{UB} - \text{LB})/\text{UB}$ where UB is the cost of the heuristic solution and LB is the lower bound given in Table III. Table IV shows, for each heuris-

TABLE II

Characteristics of the Real Instances

				Turns		Turn Penalty			
	$\|V\|$	$\|A\|$	$\|A_R\|$	Forbidden	Allowed	Right	Left	U	Straight
T1	119	180	145	5	256	2	3	5	1
T2	118	187	156	37	273	2	3	5	1
T3	109	180	140	38	278	2	3	5	1
T4	86	145	111	4	247	2	3	5	1
T5	111	171	130	10	250	2	3	5	1
V	135	265	265	69	484	1	5	11	2

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TABLE III

Computational Results

	ATSP		BK		CONS1		CONS2		CONS1 + IMP		CONS2 + IMP	
	L.B.	Time	Cost	Time	Cost	Time	Cost	Time	Cost	Time	Cost	Time
P1	21116*	12.64	21147	0.21	21137	0.28	21137	0.42	21137	0.41	21137	0.56
P2	13914*	0.77	13936	0.23	13936	0.30	13916	0.32	13930	0.53	13916	0.44
P3	15620*	1.74	15634	0.29	15634	0.33	15636	0.45	15634	0.60	15636	0.59
P4	16380*	40.74	16512	0.58	16472	0.66	16444	0.95	16428	1.39	16418	1.48
P5	22214*	3.41	22011 + 4 M	0.56	23859	0.64	22227	0.82	22711	1.63	22227	1.00
P6	26950*	27.92	26959	0.73	26958	0.85	26950*	1.22	26958	1.33	26950*	1.44
P7	24510*	19.46	24174 + 4 M	1.03	24962	1.16	24546	1.19	24896	1.99	24546	1.40
P8	20574*	12.12	19792 + 8 M	4.33	21170	4.52	20574*	4.59	21158	6.88	20574*	4.59
P9	36876	>3 h	36972	4.16	36944	4.41	36944	7.18	36944	4.90	36944	7.64
P10	18112*	12.58	17374 + 14 M	4.63	20433	5.17	18135	10.57	19640	8.38	18135	10.91
P11	36996	>3 h	37078	4.86	37088	5.08	37088	8.20	37088	5.57	37088	8.70
P12	34474	>3 h	35046	5.61	34969	5.98	34969	9.50	34815	7.61	34815	11.14
P13	26035*	25.30	24770 + 28 M	5.50	30387	6.41	26103	11.99	27441	10.13	26100	12.33
P14	35616*	37.96	35686 + M	9.73	36358	10.48	35623	13.02	36347	15.94	35623	13.58
P15	36374	>3 h	36441	9.63	36449	10.37	36449	16.24	36449	11.06	36449	16.91
P16	32471*	1335.36	32207 + 6 M	14.36	33531	15.55	32766	38.60	32936	22.82	32766	40.52
P17	42284	>3 h	42369	14.24	42363	15.39	42363	23.92	42363	16.19	42363	24.72
P18	35888	>3 h	35565 + 5 M	14.98	36541	15.80	36083	18.24	36506	24.77	36036	19.86
P19	40959	>3 h	40886 + 3 M	14.87	41784	15.84	41255	20.54	41666	25.01	41150	22.77
P20	38548*	2821.09	38339 + 3 M	16.43	38834	17.30	38564	22.12	38744	24.54	38564	22.85
P21	40216	>3 h	40338	16.40	40338	17.40	40338	27.92	40338	18.28	40338	28.71
P22	43966*	329.69	43990 + 5 M	24.46	45492	25.87	44301	74.90	44857	35.07	44176	77.28
P23	35137*	4078.31	34777 + 12 M	24.41	35953	25.79	35230	29.39	35776	37.99	35224	30.78
P24	48298*	197.63	48312 + 3 M	29.51	49147	30.59	48361	89.35	49140	43.46	48361	90.43
T1	8705*	3.04	8559 + 2 M	0.74	8708	0.82	8705*	0.99	8705*	2.13	8705*	0.99
T2	7975*	3.44	7885 + 4 M	0.79	8517	0.90	7975*	1.10	8129	1.83	7975*	1.10
T3	6142*	2.98	6042 + 3 M	1.07	6327	0.81	6142*	0.95	6244	1.76	6142*	0.95
T4	4919*	1.46	4788 + M	0.63	4922	0.44	4919*	0.50	4922	0.87	4919*	0.50
T5	7620*	2.26	7621	0.67	7621	0.71	7620*	0.80	7620*	1.42	7620*	0.80
V	6023*	1480.51	6039 + M	2.08	6185	2.27	6075	6.60	6178	7.45	6067	7.97

CPU time in seconds.

*, Optimal costs.

tic, the average and maximum deviation, whereas Table V shows the average CPU time. The deviations of the solutions generated by the BK heuristic have not been computed because most of them are not feasible. Note that all the heuristics obtain better solutions, on average, for the real instances. From these results, we may conclude that CONS2 produces much better results than CONS1. The IMP is useful to improve the solutions provided by CONS1 while it produces only slight improvements on the CONS2 solutions, which are already almost optimal. On average, the best results are obtained by CONS2 + IMP.

The CPU time used by the heuristic algorithms is negligible, so it seems that they could be useful even when applied to larger instances. In contrast, the deviation from the lower bound of the solutions provided by CONS2 + IMP is very small (0.2% on average). Nevertheless, there could exist situations where it is worthwhile to try to obtain the optimal solution using the transformation of the DRPP-TP into an ATSP. Unfortunately, the behavior of the branch-and-bound code for the ATSP is unpredictable, so we could spend a large amount of CPU time without any guarantee of finally obtaining the optimal solution.

TABLE IV

Average and Maximum Deviations

	CONS1		CONS2		CONS1 + IMP		CONS2 + IMP	
	Average	Maximum	Average	Maximum	Average	Maximum	Average	Maximum
All instances	2.30	14.32	0.25	1.41	1.35	7.78	0.20	0.97
Real instances	1.99	6.36	0.14	0.85	1.01	2.50	0.12	0.72

Deviation = 100(UB − LB)/UB.

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TABLE V
Average CPU times in seconds

	BK	CONS1	CONS2	CONS1 + IMP	CONS2 + IMP
All instances	7.59	8.07	14.75	11.39	15.43
Real instances	0.99	0.99	1.82	2.57	2.05

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